



ACADEMIC REGULATIONS & SYLLABUS

Faculty of Computer Science & Applications
Smt. Chandaben Mohanbhai Patel Institute of
Computer Applications
M.Sc.(IT) Programme
(Choice Based Credit System)



ACADEMIC REGULATIONS

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days' duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Postgraduate programme: Master of Science in Information Technology

Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)

3. Eligibility & Mode of admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from

attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6 Course Evaluation

6.1 The performance of every student in each course will be evaluated as follows:

6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 University Examination

6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

6.3 Performance at Internal & University Examination will be done on the relative grading system.

7 Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

$$SGPA = \sum C_i G_i / \sum C_i$$

where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester

$$CGPA = \sum C_i G_i / \sum C_i$$

where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

8 Detention Rule

A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in

9 Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all evaluation components in every course; and

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA ≥ 7.5 & ≤ 10
First class:	CGPA ≥ 6.0 & < 7.5
Second Class:	CGPA ≥ 5.0 & < 6.0
Pass Class:	CGPA < 5.0

11 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:
Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates

proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE): The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE): The programme specific pool of elective courses offered by respective programme.

Vision, Mission, PEOs, POs and PSOs

Vision:

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The graduates will

PEO1: Be able to understand and analyse the requirements of complex real world problems and solve them by designing optimised computer applications.

PEO2: Be able to apply skills in order to excel at professional careers.

PEO3: Be able to adopt to rapidly changing technological landscape in domains of their preference.

PEO4: Be able to act ethically and professionally while maintaining high moral values.

PROGRAM OUTCOMES (POs)

The Graduate of Computer Science and Applications will be able to :

PO1. Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.

PO2. Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.

PO3. Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies

PO4. Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well informed conclusions.

PO5. Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions

PO6. Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.

PO7. Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.

PO8. Project Management and Finance: Ability to understand, management and computing principles with computing knowledge to manage projects in multidisciplinary environments.

PO9. Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.

PO10. Societal & Environmental Concern: Ability to recognize economical, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.

PO11. Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.

PO12. Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the programme, the student should be able to

PSO1: Analyse, design and develop effective software applications for solving contemporary challenges.

PSO2: Harness the necessary skills with spirit of research and entrepreneurship.

Total Credits of Programme(4 Semesters)

108

**TEACHING SCHEME
FOR
M.Sc.IT PROGRAMME
(1ST and 2ND YEAR)

EFFECTIVE FROM
ACADEMIC YEAR 2022-23**

Teaching & Examination Scheme
Master of Science in Information Technology (M.Sc.(IT)) Programme
(Choice Based Credit System)
Effective from Year 2022 - 23
Semester - I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS120-123	Elective-I	4	-	4	4	10	20	70	-	-	-	100
MS124	Advanced Java	3	3	6	6	10	20	70	15	15	70	200
MS125	Web Development using .NET	3	3	6	6	10	20	70	15	15	70	200
MS126	Advanced Database Administration	4	3	7	7	10	20	70	15	15	70	200
HS141.02 C	Foreign Languages	-	2	2	2	-		-	30		70	100
HS105.02 C	Academic Speaking and Presentation Skills					-						
		14	11	25	25	400			400			800

Elective-I	
Course Code	Course Title
MS120	Cloud Computing Essentials
MS121	Mobile and Wireless Communications
MS122	Fuzzy Logic and Applications
MS123	Distributed Operating Systems

Teaching & Examination Scheme
Master of Science in Information Technology (M.Sc.(IT)) Programme
(Choice Based Credit System)
Effective from Year 2022 - 23
Semester - II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS223-226	Elective-II	3	3	6	6	10	20	70	15	15	70	200
MS227	Web Development Using Open Source	3	3	6	6	10	20	70	15	15	70	200
MS228	Advanced Mobile Application Development	3	3	6	6	10	20	70	15	15	70	200
MS229	Modern Software Engineering	4	-	4	4	10	20	70	-	-	-	100
HS106.02 C	Academic Writing	-	2	2	2	-	-	-	30		70	100
	University Elective-III **	-	2	2	2	-		-	30		70	100
		13	13	26	26	400			500			900

Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Internet and Web Designing and Mobile Application Development** course for others.

Elective-II	
Course Code	Course Title
MS223	Game Design
MS224	Advanced Python Programming
MS225	Block Chain Programming
MS226	HTTP Web Service for Enterprise Application

University Elective-III

No	Course Code	Course Name	Department/Faculty
1	EE782.01	Energy Audit and Management	Engineering
2	CE771.01	Project Management	Engineering
3	PT796.01	Fitness & Nutrition	Physiotherapy
4	MB651	Software based Statistical Analysis	Management
5	NR755	First Aid & Life Support	Nursing
6	OC733.01	Introduction to Polymer Science	Applied Science
7	MA771.01	Reliability and Risk Analysis	Mathematics
8	ME781.01	Occupational Health & Safety	Engineering
9	MA772.01	Design of Experiments	Mathematics
10	RD701.01	Introduction to Analytical Techniques	Applied Science
11	RD702.01	Introduction to Nanoscience And Technology	Applied Science
12	PH891	Community Pharmacy Ownership	Pharmacy
13	PH892	Intellectual Property Rights	Pharmacy
14	PSE55	Astrophysics ,Space and Cosmos	Applied Science

Teaching & Examination Scheme
Master of Science in Information Technology (M.Sc.(IT)) Programme
(Choice Based Credit System)
Effective from Year 2022 - 23
Semester - III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS328-331	Elective-III	4	3	7	7	10	20	70	15	15	70	200
MS332	Software Quality Assurance and Testing	3	3	6	6	10	20	70	15	15	70	200
MS333	Mini Project	-	12	12	12	-	-	-	100		300	400
CL144.03 C	Environmental Sciences	-	2	2	2	-	-	-	30		70	100
		07	20	27	27	200			700			900

Elective-III	
Course Code	Course Title
MS328	Advanced Game Development
MS329	Data Science
MS330	IOT enabling Technologies
MS331	Advanced web designing

Teaching & Examination Scheme
Master of Science in Information Technology (M.Sc.(IT)) Programme
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Effective from Year 2022 - 23
Semester - IV

Course Code	Course Title	Teaching Scheme			Internal	End Semester Examination		Total	
		Contact Hours		Credit	Continuous Evaluation	Report	Presentation & Viva		
		Inst.	Industry						Total
MS403	Dissertation/ Project Work	2	28	30	30	200	200	400	800

DETAILED SYLLABUS

FOR

M.Sc. (IT) PROGRAMME
(1st SEMESTER)

EFFECTIVE FROM ACADEMIC YEAR
2022-23

Teaching & Examination Scheme
Master of Science in Information Technology (M.Sc.(IT)) Programme
(Choice Based Credit System)
Effective from Year 2022 - 23
Semester - I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS120-123	Elective-I	4	-	4	4	10	20	70	-	-	-	100
MS124	Advanced Java	3	3	6	6	10	20	70	15	15	70	200
MS125	Web Development using .NET	3	3	6	6	10	20	70	15	15	70	200
MS126	Advanced Database Administration	4	3	7	7	10	20	70	15	15	70	200
HS141.02 C	Foreign Languages	-	2	2	2	-		-	30		70	100
HS105.02 C	Academic Speaking and Presentation Skills					-						
		14	11	25	25	400			400			800

Elective-I	
Course Code	Course Title
MS120	Cloud Computing Essentials
MS121	Mobile and Wireless Communications
MS122	Fuzzy Logic and Applications
MS123	Distributed Operating Systems

MS120: Cloud Computing Essentials
(100 Marks)

Contact Hours: 04

Pre-requisite: Operating System and Computer Networking concepts.

Methodology & Pedagogy: During theory lectures the emphasis will be given on the basics of cloud computing. Students will be introduced basic types, planning, service providers, mechanism, security problems and some concealed aspects of cloud computing. Students will give practical introduction in form of case study and by showing cloud infrastructure of university.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Cloud Computing	06
2	Components and Services of Cloud	09
3	Cloud Architecture	09
4	Cloud Infrastructure and Mechanisms	09
5	Cloud Security Services	08
6	Cloud Solutions and Case Studies	07

Total Hours (Theory): 48

Total Hours:48

Detail Syllabus:

Unit I : Introduction to Cloud Computing

Hours 06

History of Cloud Computing and evolution, Paradigm Shift in Computing, distributed nature of service Provisioning, Support entrepreneurship using Cloud Computing.

Unit II : Components and Services of Cloud

Hours 09

Advantages and drawbacks of Cloud Computing, Essential component for Cloud contract, Major outage of Cloud Computing and Enhancers for Cloud Computing. Introduction to SaaS, PaaS, IaaS. Introduction to Public Cloud, Private Cloud, Hybrid Cloud and Community Cloud, Storage Services for Cloud Computing.

Unit III : Cloud Architecture

Hours 09

Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture. Introduction to major Cloud Service Provider: Amazon Web Services, Google Cloud. Microsoft Windows Azure and Office 365, Hp Cloud, RackSpace, CSC Corp, Verizon Terrimark, DropBox.

Unit IV : Cloud Infrastructure and Mechanisms

Hours 09

Introduction to Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Storage Levels, Network Storage Interfaces, Object Storage Interfaces, Database Storage Interfaces, Relational Data Storage, on-Relational Data Storage, Cloud Usage Monitor, Monitoring Agent, Resource Agent, Polling Agent, Resource Replication. Introduction to Cloud Management Mechanisms: Remote Administration System, Resource Management System, SLA Management System, Billing Management System.

Unit V : Cloud Security Services

Hours 08

Introduction to Encryption, Symmetric Encryption, Asymmetric Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM), Single SignOn (SSO), Cloud-Based Security Groups. Overview of Compliance and Certification, Access Control, Organizational Control. Benefits of Business using Cloud Computing, Risk of Cloud Computing, Cost factor in Cloud Computing.

Unit VI : Cloud Solutions and Case Studies

Hours 07

Overview of Cloud services, Designing Solutions for the Cloud, Implement & Integrate Solutions, Emerging Markets and the Cloud, Tools for Building Private Cloud: IaaS using Eucalyptus, PaaS on IaaS - AppScale

Core Books:

1. Kevin L. Jackson, Scott Goessling: Architecting Cloud Computing Solutions: Packt Publication: 2018
2. Thomas Erl, Zaigham Mahmood and Ricardo Puttini: Cloud Computing Concepts, Technology & Architecture, PHI, 2013.
3. S. Srinivasan: Cloud Computing Basics, Springer, 2014.

Reference Books:

1. Derrick Rountree, Ileana Castrillo : The Basics of Cloud Computing, Syngress, 2013.
2. Rajkumar Buyya, James Broberg, Andrzej M. Goscinsk: Cloud Computing- Principles and Paradigms, John Wiley & Sons, 2011.

Web References:

1. http://whatisCloud.com/basic_concepts_and_terminology/Cloud [For basic terminology of Cloud Computing]
2. http://www.tutorialspoint.com/Cloud_Computing/ [For cloud computing lecture notes]
3. <http://www.intel.in/content/dam/www/public/us/en/documents/guides/cloudcomputingvirtualization-building-private-iaas-guide.pdf> [For cloud computing virtualization]
4. www.cs.purdue.edu/.../Any-Kim-Bhargava-MCCWorkshop.ppt [Security issues PPTs]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will learn basics of cloud computing
CO2 :	The students will be familiar with various cloud architectures and services.
CO3 :	They will get the knowledge of various network mechanics.
CO4 :	Students will get various business aspects of cloud computing with security aspects
CO5 :	They will be able to design and deploy Cloud Infrastructure

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Evolution of Cloud Computing	√				
2	Understanding Cloud Computing and basic types		√			
3	Fundamentals of Cloud Architecture and Service Providers		√			
4	Cloud Computing Mechanisms			√		
5	Cloud Computing Security and Business Use				√	
6	Hidden Aspects of Cloud Computing					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	3	1	2	2	-	-	1	-

CO2	3	1	1	1	1	1	3	1	2	2	1	3	2	-
CO3	3	2	1	2	1	3	3	2	2	2	-	3	2	-
CO4	3	3	2	3	1	3	3	2	2	2	3	-	1	-
CO5	3	3	3	3	1	3	3	3	2	2	2	-	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS 121: Mobile and Wireless Communications
(100 Marks)

Contact Hours: 04

Pre-requisite: Basic of computer and its working

Methodology & Pedagogy: During the lecture sessions, the teacher will discuss the applications of wireless communication, underlying concepts of wireless communication, cellular network and its working. The students will also learn about 5g, edge computing and security issues in wireless communication.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to wireless communications	05
2	Antenna and wave propagation	07
3	Encoding and modulation techniques	05
4	Cellular wireless network	08
5	Wireless link improvement techniques	06
6	Advanced concepts in wireless networks	05

Total Hours (Theory): 36

Total Hours: 36

Detail Syllabus:

Unit I: Introduction to wireless communications

Hours 05

Overview of communication system, component of communication system, digital vs analog communication, interface of analog to digital and digital to analog systems, conversation from analog to digital and digital to analog, examples of wireless communication, wireless vs. wired communication.

Unit II: Antenna and wave propagation

Hours 07

Radiation patterns, antenna types, propagation modes, line of sight communication, attenuation and signal distortion, free space loss, noise, atmospheric absorption, multipath refraction.

Unit III: Encoding and modulation techniques

Hours 05

Signal encoding criteria, digital data and analog signals – amplitude shift keying, frequency shift keying and phase shift keying, analog data and signal – amplitude modulation, angle modulation, analog data and digital signal – pulse code modulation, delta modulation.

Unit IV: Cellular wireless network

Hours 08

Cellular system overview, organization, frequency reuse, increasing capacity, operation of cellular system, steps in a mobile switching center (MSC) controlled call between mobile users, mobile radio propagation effect, additional function in MSC controlled call, handoff performance metrics, handoff strategies used to determine instance of handoff, power control, traffic engineering.

Unit V: Wireless link improvement techniques

Hours 06

Equalization, diversity error detection methods – parity check, cyclic redundancy check, block error correction code – block codes, hamming codes, cyclic codes, reed Solomon codes.

Unit VI: Advanced concepts in wireless concepts

Hours 05

Introduction to Wi-Fi, WiMAX, ZigBee Networks, Wireless Adhoc Network, Overview of 4g and 5g cellular technologies, edge computing and fog computing, Security aspects of wireless computing.

Core Books:

1. William Stallings: Wireless communication & networks: 3rd Edition: Pearson: 2012.
2. David Tse: Fundamentals of Wireless Communication: 2nd Edition: Cambridge Uni. Press: 2014
3. Aditya K. Jagannatham: Principles of modern wireless communication Systems Theory and Practice: 1st Edition: McGraw Hill: 2017

Reference Books:

1. T.S. Rappaport, "Wireless Communications Principles and Practice", 2nd edition, PHI, 2002.
2. William C.Y. Lee, "Mobile Cellular Telecommunications Analog and Digital Systems", 2nd edition, TMH, 1995

Web References:

1. https://www.tutorialspoint.com/wireless_communication/index.htm [Wireless networking tutorials]
2. <https://www.javatpoint.com/mobile-communication-tutorial> [Mobile computing tutorials]
3. <https://www.youtube.com/watch?v=f2wIHL1Sok8> [Cellular system and wireless concepts -NPTEL course YouTube link]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Student will understand role of wireless networks in communication system.
CO2 :	Student will get insight into underlying principles of wireless communication system.
CO3 :	The student will grasp concepts of error and flow control of wireless network.
CO4 :	The student will get awareness of cellular network system and its working.
CO5 :	The student will know about modern and upcoming wireless communication technologies and concepts along with security aspects.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to wireless communications	√				
2	Antenna and wave propagation		√			
3	Encoding and modulation techniques			√		
4	Cellular wireless network				√	
5	Wireless link improvement techniques				√	
6	Advanced concepts in wireless concepts					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	-	1	1	3	2
CO2	3	3	3	3	3	3	3	2	3	-	2	2	3	2
CO3	2	2	3	3	2	2	3	2	2	-	1	1	3	2
CO4	3	3	3	3	3	3	3	2	2	2	3	3	3	3
CO5	3	3	3	3	3	3	3	3	2	1	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS122-Fuzzy Logic and Applications
(100 Marks)

Contact Hours: 04

Pre-requisite: Basics of Set Theory, Fundamental of Computing

Methodology & Pedagogy: During theory lectures Basics of Fuzzy Logic, Fuzzy Set Theory, Fuzzy Relations, Fuzzification, Fuzzy systems and Applications will be introduced to students. Students will be made familiar with applications and systems related to Fuzzy Theory. Appropriate case studies will be discussed and assigned to students for real exposure related to Fuzzy logic.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Fuzzy Logic	07
2	Classical Sets and Fuzzy Sets	09
3	Classical Relations and Fuzzy Relations	08
4	Fuzzification and Defuzzification	07
5	Fuzzy Systems and Classification	09
6	Fuzzy Logic - Applications	08

Total Hours (Theory): 48

Total Hours: 48

Detail Syllabus:

Unit I : Introduction to Fuzzy Logic

Hours 07

A Historical Perspective, The Utility of Fuzzy Systems, imitations of Fuzzy Systems, The Illusion: Ignoring, Uncertainty and Accuracy, Uncertainty and Information, The Unknown, Fuzzy Sets and Membership, Chance Versus Fuzziness, Sets as Points in Hypercubes.

Unit II : Fuzzy Set Theory

Hours 09

Classical Sets : Operations on Classical Sets, Properties of Classical (Crisp) Sets, Mapping of Classical Sets to Functions, Fuzzy Sets: Fuzzy Set Operations, Properties of Fuzzy Sets, Alternative Fuzzy Set Operations, Fuzzy Arithmetic.

Unit III : Classical Relations and Fuzzy Relations

Hours 08

Crisp Relations: Cardinality of Crisp Relations, Operations on Crisp Relations, Properties of Crisp Relations, Composition.

Fuzzy Relations: Cardinality of Fuzzy Relations, Operations on Fuzzy Relations, Properties of Fuzzy Relations, Fuzzy Cartesian Product and Composition, Fuzzy Intervals.

Unit IV : Fuzzification and Defuzzification

Hours 07

Features of the Membership Function, Various Forms, Fuzzification, Defuzzification to Crisp Sets and scalars, Approximate Reasoning, Fuzzy Inference, Fuzzy Logic in Databases, Information Retrieval with Fuzzy Logic, Quantification, Activity for choosing a suitable alternative needed for realizing goal.

Unit V : Fuzzy Systems and Classification

Hours 09

Fuzzy Logic - Control System, Adaptive Fuzzy Controller: Basic Steps for Implementing Adaptive Algorithm, Parameters for selecting an Adaptive Fuzzy Controller.

Fuzzy (Rule-Based) Systems, Graphical Techniques of Inference, Classification by Equivalence Relations, Crisp Relations, Fuzzy Relations, Fuzzy Classification: Hard c-Means (HCM), Fuzzy c-Means (FCM)

Unit VI : Fuzzy Logic - Applications

Hours 08

Fuzzy Logic - Applications in Pattern Recognition, Classification, Computer Network, Medical, Knowledge Engineering and Data Mining. Fuzzy logic with Neural Network. Intelligent control of complex systems. Exercises on Fuzzy Logic.

Core Books:

4. Timothy J. Ross: Fuzzy logic with Engineering Applications : Edition (4th), John Wiley & Sons, Ltd., 2017
5. Hans-Jurgen Zimmermann: Fuzzy Set Theory - and its Applications: Edition (4th), Springer Netherlands, 2001

Reference Books:

1. Klir, G., Yuan, B. : Fuzzy Sets and Fuzzy Logic: Theory and Applications : Edition (4th), Pearson Education, 2015.
2. G. Chen and T. Pham : Introduction to fuzzy sets, fuzzy logic & Fuzzy control systems CRC Press, 2000.
- 3.
4. Hung T. Nguyen, Elbert A. Walker: A First Course in Fuzzy Logic, CRC Press, 2005.
5. A. K. BHARGAVA: Fuzzy Set Theory Fuzzy Logic and their Applications S CHAND & Company Limited, 2013.
6. M. GANESH: INTRODUCTION TO FUZZY SETS AND FUZZY LOGIC, PHI Learning, 2006.
7. Barnabas Bede: Mathematics of Fuzzy Sets and Fuzzy Logic, Springer Berlin Heidelberg, 2012.

Web References:

1. <https://plato.stanford.edu/entries/logic-fuzzy/> [Fuzzy Logic]
2. <https://www.mathworks.com/help/fuzzy/what-is-fuzzy-logic.html> [What Is Fuzzy Logic?]
3. <https://www.guru99.com/what-is-fuzzy-logic.html> [Fuzzy Logic Tutorial: What is, Architecture, Application, Example]

4. https://www.tutorialspoint.com/fuzzy_logic/fuzzy_logic_applications.htm [Fuzzy Logic - Applications]

Course Outcomes: Upon successful completion of the course, students will be able

CO1 :	To understand concept of fuzzy logic principles.
CO2 :	To understand the skill in basic understanding of fuzzy logic.
CO3 :	To explore the components of fuzzy set and fuzzy operations.
CO4 :	To explore the functional components of fuzzy logic classification and controller.
CO5 :	To explore applications related to fuzzy logic.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Fuzzy Logic	√				
2	Classical Sets and Fuzzy Sets		√	√		
3	Classical Relations and Fuzzy Relations			√	√	
4	Fuzzification and Defuzzification			√	√	
5	Fuzzy Systems and Classification				√	√
6	Fuzzy Logic - Applications					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	-	-	3	-	-	-	-	2	3	2
CO2	3	3	2	2	-	-	3	-	-	-	-	2	3	2
CO3	3	3	2	2	1	-	3	-	-	-	-	2	3	2
CO4	3	3	2	2	1	-	3	-	-	-	-	2	3	2
CO5	2	2	2	3	2	-	3	-	-	-	-	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS123: Distributed Operating Systems
(100 Marks)

Contact Hours: 04

Pre-requisite: Operating Systems concepts

Methodology & Pedagogy: During theory lectures fundamental concept of Distributed Systems will be covered along with the issues and management of distributed memory like synchronization in Distributed Operating Systems, Resource Management, Handling file and database in distributed systems and the Security concerns.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Overview of Distributed Systems	08
2	Distributed Shared Memory	09
3	Synchronization in Distributed Operating Systems	09
4	Process and Resource Management in Distributed Operating Systems	08
5	Distributed File Systems	07
6	Distributed Web based System and Security	07

Total Hours (Theory): 48

Total: 48

Detailed Syllabus:

Unit – I: Overview of Distributed Systems

Hours: 08

Computer architecture: CICS, RISC, Multi-core, Computer networking: ISO/OSI Model, Evolution of operating systems, Introduction to Distributed operating system, issues in designing distributed operating system – transparency, Reliability, Flexibility, performance, scalability, heterogeneity, security, Emulation of existing operating system, Introduction to distributed computing environment, Introduction to distributed computing systems (DCS), DCS design goals, Transparencies, Fundamental issues.

Unit – II: Distributed Shared Memory

Hours: 09

Introduction to RPC, RPC Model, Implementing RPC Mechanism, Stub Generation, RPC Message, Introduction to Distributed Shared Memory, General Architecture of DSM Systems, Design and implementation issues of DSM, Granularity, Structure of Shared memory space, Message passing communication, Transaction Communication, Group communication; Broadcast atomic protocols.

Unit – III: Synchronization in Distributed Operating Systems

Hours: 09

Distributed Coordination, Temporal ordering of events, Lamport's logical clocks, Vector clocks; Ordering of messages, Physical clocks, Global state detection Clock Synchronization – Implementation, drifting of clocks, Clock synchronization issues, Clock Synchronization algorithms, Mutual Exclusion – Centralized approach, Distributed Approach, Token Passing Approach, Deadlocks in distributed systems ,Load scheduling and balancing techniques.

Unit – IV: Process and Resource Management in Distributed Operating Systems

Hours: 08

Introduction to resource management in distributed operating system, Desirable features of good global scheduling algorithm, Task Assignment approach, Load balancing Approach, Load Sharing Approach, Process migration – desirable feature of Process migration, process migration mechanisms, Process migration in heterogeneous systems, Advantages of process migration.

Unit – V: Distributed File and Database Systems

Hours: 07

What is Distributed File system – Remote information sharing, User mobility, Availability, Diskless workstation. Types of services in distributed file system – Storage service, True file Service, Name Service. Desirable features of Distributed file system, File Models, File accessing models, Distributed database system.

Unit – VI: Distributed Web based System and Security

Hours: 07

Architecture, Processes, Communication, Naming, Synchronization, Consistency and Replication: Web Proxy Caching, Replication for Web Hosting Systems, Replication of Web Applications, Introduction of Security in Distributed OS, Overview of security techniques, Features, Need, Access Control, Security Management.

Core Books:

1. Tim Kindberg, George Coulouris, Jean Dollimore : Distributed Systems : Concepts & Design Paperback Pearson Education,2010.
2. Pradip K. Sinha: Distributed Operating Systems Concepts and Design, Eastern Economy Edition, PHI, 2007.
3. Mukesh Singhal, Niranjana Shivaratri : Advanced Concepts In Operating Systems Paperback, McGraw Hill Education,2017.

Reference Books:

1. Andrew S. Tanenbaum: Distributed Operating Systems, 5th Edition, Pearson, 2008.
2. Doreen L. Galli: Distributed Operating Systems Concepts and Practice, Prentice Hall, 2000.
3. Andrew S. Tanenbaum, Herbert Bos: Modern Operating Systems, 4th Edition, Pearson Education, 2014.

Web References:

1. www.geeksforgeeks.org/types-of-operating-systems/ [Overview of Distributed OS]
2. www.javatpoint.com/distributed-operating-system [Examples and application of Distributed OS]

3. www.tutorialspoint.com/Distributed-Database-Management-System [Database Management of Distributed OS]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be familiar with Distributed Operating Systems and Shared Memory.
 C02 : Be familiar with the Synchronization issues in Distributed Operating Systems
 C03 : Be familiar with the Process and Resource management.
 C04 : Understand working of file and database in the distributed environment.
 C05 : Understand the working of web based system and security concerns in the distributed systems.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Overview of Distributed Systems	✓				
2	Distributed Shared Memory and Remote Procedure Calls	✓				
3	Synchronization in Distributed Operating Systems		✓			
4	Process and Resource Management in Distributed Operating Systems			✓		
5	Distributed File and Database Systems				✓	
6	Distributed Web based System and Security					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C01	2	2	2	2	2	2	2	2	2	2	2	2	2	1
C02	2	3	2	3	3	2	2	2	2	2	2	2	2	1
C03	2	3	2	3	3	2	2	2	2	2	2	2	2	2
C04	3	2	2	2	3	2	2	2	2	2	2	2	2	2
C05	3	2	2	2	3	2	2	2	2	2	2	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS124: Advanced Java

(200 Marks)

Contact Hours: 06

Pre-requisite: Fundamental Knowledge of Core Java (Java SE) concepts.

Methodology & Pedagogy: Advanced Java fundamentals like JDBC, Servlet, JSP and Servlet Security provides capabilities and features for developing web application which is essential for developer in order to understand entire web application development life cycle. During the theory sessions of the syllabus, fundamentals of the mentioned technologies will be explained in detail with real time examples. Moreover, practical session will provide hands-on exposure about JDBC, Servlet, JSP and Servlet Security with variety of practical assignment. Also, MVC architecture using these technologies will be covered during theory as well as practical sessions.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	JDBC Programming	06	36
2	Web Application Components Fundamentals	05	
3	Servlet API	08	
4	Advanced Servlet Features and Security	05	
5	Java Server Pages	07	
6	Packaging and Deploying Web Applications	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: JDBC Programming

Hours 06

The JDBC Connectivity Model, Database Programming: Connecting to the Database, creating a SQL Query, Getting the Results, Updating Database Data, Error Checking and the SQLException Class, the SQLWarning class, the Statement Interface, PreparedStatement, CallableStatement, the ResultSet Interface, Updatable Result sets, JDBC Types, Executing SQL Queries, ResultSetMetaData, Executing SQL Updates, Transaction Management

Unit II: Web Application Components Fundamentals

Hours 05

Basics of web application development and web components, Introduction of MVC design pattern, Fundamentals of: Containers, Packaging Web Applications, Web Application Structure, JAR Files, WAR Files, HTTP, HTTP request methods, HTML Form Processing, HTTP request response cycle.

Unit III: Servlet API

Hours 08

Servlet Model: Introducing Servlet, Introducing Servlet & the MVC Pattern, Introducing javax.servlet Packages, Introducing HTTP & Servlets, Understanding the Request/ Response Cycle, Input & Output Streams, Introducing Servlet/ Container Communication, Introducing ServletContext and ServletConfig, RequestDispatcher Interface.

The Filter API: Filter, Filter Chain and Filter Config

Unit IV: Advanced Servlet Features and Security

Hours 05

Cookies and Session Management: Understanding state and session, Understanding Session Timeout and Session Tracking, URL Rewriting.

Understanding the Stateless nature of HTTP, Why Track Client Identity & State? Maintain Sessions, Session Management Using the Servlet API.

Unit V: Java Server Pages

Hours 07

JSP Overview: The Problem with Servlets, Life Cycle of JSP Page, JSP Processing, JSP Application Design with MVC, Setting Up the JSP Environment

JSP Directives, JSP Action, JSP Implicit Objects JSP Form Processing, JSP Session and Cookies Handling, JSP Session Tracking

JSP Database Access, JSP Standard Tag Libraries, JSP Custom Tag, JSP Expression Language, JSP Exception Handling, JSP XML Processing

Unit VI: Packaging and Deploying Web Applications

Hours 05

Basics of server container, build a WAR artifact for the web application, configure the Docker connection node and deploy the application to server container

Core Books:

1. Cay S Horstmann, Gary Cornell: Core Java, Volume II – Advanced Features, 8th Edition, Pearson Education.

2. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 2, Advanced Technologies, 2nd Edition, Pearson Education, 2008.

Reference Books:

1. Bryan Basham, Kathy Sierra and Bert Bates: Head First Servlet and JSP, O'Reilly Publication, 1st Edition
2. James Keogh: Complete Reference J2EE, Mcgraw publication

Web References:

1. <http://courses.coreservlets.com/Course-Materials/csajsp2.html> [Servlet Basics]
2. <http://www.ceit.es/asignaturas/IntelInfo/Recursos/Servlets/JavaServlets.pdf> [Servlet Tutorial PDF]
3. <http://www.msuniv.ac.in/AdvancedJavaProgrammingwithDatabaseApplication.pdf> [JDBC Tutorial]
4. www.doc.ic.ac.uk/~rcheung/teaching/2720/ppt/lecture12.ppt [JSP Tutorial Slides]

Course Outcomes: Upon successful completion of the course, students will:

CO1 : Able to connect Java application with various RDBMS using JDBC

CO2 :	Have understanding of web application architecture and terminologies
CO3 :	Able to develop real time web application using Servlet and JSP technologies
CO4 :	Be familiar with the concepts of JSPs, Servlets and related security issues.
CO5 :	Apply Model-View-Controller architecture to build complex client-server applications

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	JDBC Programming	√		√		
2	Web Application Components Fundamentals		√	√		
3	Servlet API			√	√	√
4	Advanced Servlet Features and Security			√	√	√
5	Java Server Pages			√	√	√
6	Packaging and Deploying Web Applications					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	1	1	-	-	-	-	-	-	-	3	2
CO2	3	2	3	2	-	-	-	-	-	-	2	1	2	2
CO3	3	2	3	2	2	-	-	-	-	2	2	3	3	3
CO4	3	2	3	2	2	2	-	-	1	-	2	1	3	2
CO5	2	2	3	3	2	-	1	1	-	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS125: Web Development Using .NET
(200 Marks)

Contact Hours: 06

Pre-requisite: Familiarity with basic concepts of object oriented programming

Methodology & Pedagogy: In the lectures students will learn about basics of .NET framework, ASP.NET and other advanced concepts related to programming with .NET framework. In the lab sessions the student will learn about solving real world problems with .NET framework and related technologies.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to ASP.NET	05	36
2	Applications development with ASP.NET	06	
3	Database with ADO.Net	08	
4	MVC architecture in .NET framework	08	
5	SOA in .NET framework	05	
6	AJAX in .NET	04	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: Introduction to ASP.NET

Hours 05

Introduction to .NET framework, Introduction to Web Programming, Introduction to ASP and ASP.NET, Deploying ASP.NET Application

Unit II: Applications development with ASP.NET

Hours 06

ASP.NET Page Life Cycle, Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config, Developing a web form, Working with master pages, State management techniques, Application Tracing, Error Handling

Unit III: Database with ADO.Net

Hours 08

Introduction to ADO and ADO.NET, ADO.NET architecture, Connection Oriented and Connection Less architecture, Binding data to web controls and working with data controls, working with XML

Unit IV: MVC architecture in .NET framework

Hours 08

Introduction to MVC Architecture in ASP.NET, CRUD operation using MVC architecture, working with web controls in MVC architecture, introduction to LINQ.

Unit V: SOA in .NET framework

Hours 05

Introduction to service oriented architecture concept, SOAP protocol, Concept of UDDI and WSDL, Applications of SOA, Building SOA application with ASP.NET.

Unit VI: AJAX in .NET

Hours 04

Introduction to ASP.NET AJAX, AJAX control toolkit and extender controls.

Core Books:

6. Cogent Solutions Inc.: ASP.Net 4.5 Black book, Dreamtech press, 2019
7. Mridila Parihar, Essam Ahmed: ASP .Net Bible, Wiley, 2017
8. Stephon Walther: ASP.Net Unleashed, BPB publication, 2018

Reference Books:

1. Mesbah Ahmed, Chris Garrett, Jeremy Faircloth, Chris Payne: ASP.Net Programming. Developer's Guide, Dreamtech, 2016.
2. Greg Buczek: ASP.Net Tips & Techniques, Tata McGraw Hill Edition, 2016.
3. Bolton, Justin Langford, Glenn Berry, Gavin Payne, Amit Banerjee, Rob Farley: Professional SQL Server 2019 internals and troubleshooting, Wiley India publication, October, 2019.

Web References:

4. <http://msdn.microsoft.com/en-us/aa336522.aspx> [For software download]
5. <https://docs.microsoft.com/en-us/aspnet/overview> [For ASP.NET technical documentation]
6. https://www.tutorialspoint.com/dotnet_core/dotnet_core_overview.htm [for .NET Core]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will be able to understand .NET framework and develop ASP.NET application.
CO2 :	Students will be able to develop database centric applications.
CO3 :	Students will be familiar with concepts of MVC for application development.
CO4 :	Students will able to understand and develop SOA based applications with .NET framework.
CO5 :	Students will be familiar with the concepts of AJAX

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to ASP.NET	√				
2	Applications development with ASP.NET	√				
3	Database with ADO.Net		√			
4	MVC architecture in .NET framework			√		
5	SOA in .NET framework				√	
6	AJAX in .NET					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	-	2	1	-	-	2	-	-	2	1
CO2	2	3	3	2	-	2	1	-	-	2	2	1	3	1
CO3	1	2	2	-	-	-	-	-	-	-	2	1	3	2
CO4	2	3	3	-	2	-	-	-	-	-	2	1	3	2
CO5	3	2	3	2	-	-	3	-	-	-	-	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS126: Advanced Database Administration
(200 Marks)

Contact Hours: 07

Pre-requisite: To be able to implement a simple DBMS and design and implement information systems using DBMS technology.

Methodology & Pedagogy: During theory lectures, illustrations of certain complex real world applications, which emphasize the use of advanced concepts of databases, will be discussed. The fundamental concepts regarding database development activities, various database management systems and other advanced issues in database management will also be discussed. In addition, there may be announced or unannounced quizzes/assignments. During Practical sessions, students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to DBA	6	36
2	User Access and Security	9	
3	Data Storage management & Migration	9	
4	Managing Database Backup and Recovery	9	
5	Database Security and Auditing	7	
6	Query Processing and Optimization	8	

Total Hours (Theory): 48

Total Hours (Lab) : 36

Total Hours: 84

Detail Syllabus:

Unit I: Introduction to DBA

Hours 06

Types of Oracle Database Users, DBA Roles and Responsibilities; Database Architecture; ORACLE logical and physical database structure; Memory and Process Structure, SQLPLUS Overview, creating a database.

,

Unit II: User Access and Security

Hours 09

Administrative User Accounts, Database Administrator Authentication, Administrative Privileges and its operations, managing Password authentication, Granting and Revoking Administrative Privileges

Unit III: Data Storage management and Migration

Hours 09

Disk Storage, Basic File structure and Indexing, Indexing structure for Files, Data Migration

Data Utilities (SQL loader and Import – Export)

Unit IV : Managing Database Back-up and Recovery

Hours 09

Backup and Recovery Overview, Database backup, restoration and recovery, defining a backup and recovery strategy, Backup and Recovery options; Data Dump; User-Managed Backup and Recovery; Configuring RMAN; RMAN Backups, Restore and Recovery.

Unit V : Database Security and Auditing

Hours 07

Database Security and Auditing; Database Authentication Methods; Database Authorization Methods; Data Encryption Techniques, Virtual Private Database; Managing Users and Security: Profiles, managing users, managing privileges, managing roles

Unit: VI : Query Processing and Optimisation

Hours 08

Notation for Query tree and Graphs, Heuristic optimization of Query tree, General rules for transformation, Converting Query trees into Query Execution Plans, Cost component for Query tree and its optimization

Core Books:

1. Oracle® Database Database Administrator's Guide by Mark Doran, Padmaja Potineni, Rajesh Bhatiya, Oracle Press.

Reference Books:

1. Abraham Silberschatz, Henry F.Koeth, S.Sudarshan: Database System Concepts, 6th edition, McGraw Hill Publication.
2. Ramez Elmasri,Shamkant B.Navathe:Fundamentals of Database Systems, 5th Edition, Pearson Publication.
3. C.J. Date, An Introduction to Database Systems (eighth edition), Addison Wesley, 2000

Web References:

[A Practical Guide to Oracle Database Administration for DBA & Developers \(oracletutorial.com\)](http://oracletutorial.com)

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Create and manage databases
CO2 :	Be familiar with database storage management
CO3 :	Manage backup and recovery
CO4 :	Control user security
CO5 :	Managing database query optimization

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to DBA	√				
2	User Access and Security	√	√			
3	Data Storage management & Migration	√	√	√		
4	Managing Database Backup and Recovery		√		√	√
5	Database Security and Auditing			√		√
6	Query Processing and Optimization				√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	3	1	3	1	2	-	-	-	-	3	-
CO2	2	2	3	2	1	3	3	2	-	-	-	-	3	-
CO3	3	3	2	1	2	3	2	1	-	-	-	-	3	3
CO4	3	3	3	1	2	3	3	1	-	-	-	-	3	3
CO5	3	3	2	3	2	2	3	1	-	-	-	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS141.02 A: FOREIGN LANGUAGES (French)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/01	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- French Language Studies- Introduction (Coursera)

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to French Language	08
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06

Total hours (Theory):--

Total hours (Practical) :30

Total hours :30

Detailed Syllabus:

1.	Introduction to French Language	08 Hours	28%
	Facts and figures about French Language; Basic French Linguistics-* Alphabets * Accents * Liaison * Nasalization French Culture, Differ between French and English; Grammar- Subject Pronoun, Verbs: (être, avoir, habiter, regarder, manger ... “er” verb), Form of address, Numbers		

	(1 to 20), Nouns and plurals of nouns, The expression: C'est, Il y a; Presentation: -1) Self-Introduction-2) Question and answering; Dialogue		
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08 Hours	28%
	Grammar -Definite articles, Indefinite articles, Present tense (Positive Forms, Negative Forms), Numbers (21 to 100, 100-1000), Days, Months, Family, Verbs: (aller, venir, finir, pouvoir, vouloir ... "ir" verb); Social Links -1), My family & relations 2) Appointments 3) Gathering information from someone; Dialogue		
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08 Hours	28%
	Grammar - Common Adjectives, Comparative Adjectives, Common Adverbs, Interrogative Forms, The expression: "On", Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs: (Prendre, Apprendre, Comprendre, faire ... "re" verb); Work , Study and Travel -1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport); Dialogue		
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06 Hours	26%
	Grammar -1) Common Prepositions 2) Common Conjunctions 3) Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: ("er", "ir", "re" etc...); Food & Shopping -1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/At Restaurant / At Market); Dialogue		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Gain basic communication skills in French language with preliminary understanding of grammar
CO2	Develop vocabulary required to speak about him/herself and his/her immediate environment.
CO3	Become capable of interacting in simple ways, to ask simple questions to get necessary information, to reply simple questions.
CO4	Become capable of understanding and using simple instructions in their personal, academic and professional environments.
CO5	Develop skills and intelligences to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new global trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO6	-	-	-	-	-	-	3	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- **Text book:**

1. Complete French: All-In-One, McGraw-Hill, Amazon
2. Best for Grammar: Easy French Step-by-Step, McGraw-Hill, Amazon

- **Reference book:**

1. Basic French: McGraw-Hill, Amazon
2. French Grammar for Beginners, Amazon

- **Web material:**

1. <https://alison.com/course/french-language-studies-introduction>
2. <https://alison.com/course/basic-french-language-skills-for-everyday-life-revised-2017>
3. <http://www.bbc.co.uk/languages/french/>
4. <https://www.loecsen.com/en/learn-french>
5. <https://www.youtube.com/watch?v=ujDtm0hZyII>

HS105.02 C: ACADEMIC SPEAKING AND PRESENTATION SKILLS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Beginner/Intermediate level language proficiency

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Foundations of Advance Communication	04
2.	Art of Conversation	06
3.	Science of Power Speaking	06
4.	Academic Speaking Application – Part I	08
5.	Academic Speaking Application – Part II	06

Total hours (Theory): --

Total hours (Practical) :30

Total hours :30

Detailed Syllabus:

1.	Foundations of Advance Communication	04 Hours	14%
	Meaning and Definition of Advance Communication; Advance Communication in Digital, Social, Mobile World; Strategies for Advance Communication; Meaning and Concept of Academic Language; High Frequency Academic Vocabulary		
2.	Art of Conversation	06 Hours	20%

	Describing people, places and things; Expressing opinions; Making suggestions; Persuading someone; Interpreting and Summarizing		
3.	Science of Power Speaking	06 Hours	20%
	Phonemes, Word Stress, Pronunciation, Intonation, Pause, Register, Fluency, Prosody, Lexical Range		
4.	Academic Speaking Application – Part I	08 Hours	26%
	Art of Oratory, Formal Presentation, Speech Analysis – Decoding Best Speeches		
5.	Academic Speaking Application – Part II	06 Hours	20%
	Job Interview, Group Discussion, Meeting		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	understand and demonstrate advance communication and academic speaking skills
CO2	demonstrate ability to communicate in diverse situations
CO3	activate and extend their linguistic and communicative competence
CO4	demonstrate the formal presentation skills
CO5	demonstrate performing ability at group discussion and personal interview

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	2	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	3	-	-	-

Recommended Study Material:

Reference book:

1. *Business Communication Today* (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina

2. *Effective Speaking Skills* by Terry O' Brien
3. *Speak Better Write Better* by Norman Lewis
4. *Well Spoken: Teaching Speaking to All Students* by Erik Palmer
5. *Let Us Hear Them Speak : Developing Speaking – Listening Skills in English* by Jayshree Mohanraj
(Publisher – Sage Publication)
6. *The craft of scientific presentations: Critical steps to succeed and critical errors to avoid.* New York:
Springer by Michael Alley
7. *Presentation Skills in English* by Bob Dignen (Publisher: Orient Black Swan)

Web material:

1. *TED Talk : How to speak so that people want to listen*
2. https://www.ted.com/talks/julian_treasure_how_to_speak_so_that_people_want_to_listen?language=en
3. *TED Talk: The 110 techniques of communication and public speaking*
4. https://www.ted.com/talks/david_jp_phillips_the_110_techniques_of_communication_and_public_speaking

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**M.Sc. (IT) PROGRAMME
(2nd SEMESTER)**

EFFECTIVE FROM ACADEMIC YEAR 2022-23

Teaching & Examination Scheme
Master of Science in Information Technology (M.Sc.(IT)) Programme
(Choice Based Credit System)
Effective from Year 2022 - 23
Semester - II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS223-226	Elective-II	3	3	6	6	10	20	70	15	15	70	200
MS227	Web Development Using Open Source	3	3	6	6	10	20	70	15	15	70	200
MS228	Advanced Mobile Application Development	3	3	6	6	10	20	70	15	15	70	200
MS229	Modern Software Engineering	4	-	4	4	10	20	70	-	-	-	100
HS106.02 C	Academic Writing	-	2	2	2	-	-	-	30		70	100
	University Elective-III **	-	2	2	2	-		-	30		70	100
		13	13	26	26	400			500			900

Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Internet and Web Designing and Mobile Application Development** course for others.

Elective-II	
Course Code	Course Title
MS223	Game Design
MS224	Advanced Python Programming
MS225	Block Chain Programming
MS226	HTTP Web Service for Enterprise Application

University Elective-III

No	Course Code	Course Name	Department/Faculty
1	EE782.01	Energy Audit and Management	Engineering
2	CE771.01	Project Management	Engineering
3	PT796.01	Fitness & Nutrition	Physiotherapy
4	MB651	Software based Statistical Analysis	Management
5	NR755	First Aid & Life Support	Nursing
6	OC733.01	Introduction to Polymer Science	Applied Science
7	MA771.01	Reliability and Risk Analysis	Mathematics
8	ME781.01	Occupational Health & Safety	Engineering
9	MA772.01	Design of Experiments	Mathematics
10	RD701.01	Introduction to Analytical Techniques	Applied Science
11	RD702.01	Introduction to Nanoscience And Technology	Applied Science
12	PH891	Community Pharmacy Ownership	Pharmacy
13	PH892	Intellectual Property Rights	Pharmacy
14	PSE55	Astrophysics, Space and Cosmos	Applied Science

MS223: Game Design
(200 Marks)

Contact Hours: 06

Pre-requisite: Knowledge of C++, physics and mathematics concepts

Methodology & Pedagogy: During theory lectures illustrations Graphics, animation and various concepts regarding Game Development. Emphasize will be given on some mathematical and physics concepts, Fundamental of Graphics and Graphics Programming, Vectors, Image Data, 2D graphics, Collision detections, Fundamental of Game programming, Gameloop, Game Engine and many more. During Practical sessions, students will be developing Simple 2D graphics, Implement animation on that graphics, Handles Input and Output of data according to event generation. Student will develop simple 2D game using Visual C++.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Basics of Graphics and Game	04	36
2	Important Aspects of Mathematics and Physics	07	
3	Graphics Programming Fundamentals	05	
4	Animation, Transformations and Collision Detection	07	
5	Actual Game Programming	09	
6	Game Engine Introduction	04	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: Basics of Graphics and Game

Hours 04

The World of Computer Graphics, Current and Future Application Areas, User-Interface Considerations, History of Games and Game Development.

Unit II: Important Aspects of Mathematics and Physics

Hours 07

Vector Addition, Subtraction, Dot Product, Cross Product, Magnitude, Finding the Normal of a Vector, Projection of one vector over another vector, Matrix and operations: Matrix Addition, Subtraction, Multiplication, Transpose, and Inverse of a Matrix, Basic Trigonometry: Meaning of Sin, Cosine, and Tan functions, Cartesian coordinate system, Polar Coordinate System, Nested Coordinate Systems Physics Concepts: Speed, Velocity, Acceleration, Mass, Force, Gravity, Friction.

Unit III: Graphics Programming Fundamentals

Hours 05

Video Ram, Pixels and screen sizes, Graphics Card, Frame Buffer, Double Buffer, Screen, Refresh Rates (on LCD display units), Screen Resolutions, Vector Graphics and Raster Graphics, Image Data - How are images represented in a Bitmap file, Drawing images on a screen.

Unit IV: Animation, Transformations and Collision Detection

Hours 07

Programming Simple Animations and understanding animation frame rates, Concept of Sprites in Animation, Line Drawing Algorithm, Circle Drawing Algorithm, Transformations - Translate, Rotate, Scale, Transformation Matrix, Collision detection techniques for 2D Shapes: Circle to Circle, Rectangle to Rectangle, Circle to Line, Circle to Rectangle, Line to Line.

Unit V: Actual Game Programming

Hours 09

Application and Window Management, Input Management, File Input and Output, Audio, Programming, Game loop, Frame rate independent entity behaviors.

Unit VI: Game Engine Introduction

Hours 04

Introduction to Game Engine, Types of Game Engines, Some of the most popular Game Engines of today: Unity, Unreal, Cocos2d-X, Box2d.

Core Books:

1. Laszlo Michael, Computational Geometry and Computer Graphics in C++, Pearson Education; First edition, 2017.
2. Roger Mayne, Introduction to Windows and Graphics Programming with Visual C++.NET, World Scientific Publishing Co. Pte. Ltd., 2005.
3. John Horton, Beginning C++ Game Programming, Packt Publishing Limited, 2016.
4. Michael Dawson, Beginning C++ Through Game Programming, Cengage; 4th edition, 2014.
5. Eric Lengyel, Foundations of Game Engine Development, Volume 1: Mathematics, Terathon Software LLC; 1 edition, 2017.

Reference Books:

4. John F. Hughes, Andries van Dam, Morgan McGuire, David F. Sklar, James D. Foley, Steven K. Feiner, Kurt Akeley, "Computer Graphics: Principles and Practice in C", 3rd Edition, Addison Wesley Professional, 2013
5. David Conger, Ron Little, Creating Games in C++: A Step-by-step Guide, Pearson Education, 2006.
6. Fabien Sanglard, Game Engine Black Book: Wolfenstein 3D, Create Space Independent Publishing Platform; 1 edition
7. Constantine Pozrikidis, Introduction to C++ Programming and Graphics, Springer-Verlag New York Inc.; 2007
8. James M. Van Verth, Lars M. Bishop, Essential Mathematics for Games and Interactive Applications, A K Peters/CRC Press; 3rd edition, 2015

Web References:

1. <http://cplusplus.happycodings.com/computer-graphics/> [Graphics Tutorial]
2. <http://cplusplus.happycodings.com/games/> [Game Tutorial]
3. <https://www.geeksforgeeks.org/basic-graphic-programming-in-c/>
4. <https://www.thecrazyprogrammer.com/2013/04/simple-program-to-create-moving-car-in.html>
5. <http://mytechnotrick.blogspot.com/2015/07/#>
6. <https://www.youtube.com/watch?v=hfBR6ULOOjw>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be having basic knowledge of computer graphics
CO2 :	Be familiar with tools and techniques used for graphics and game programming
CO3 :	Be able to perform mathematic calculation required for game programming
CO4 :	Be familiar with the various image data and shapes.
CO5 :	Be able to create animated and interactive graphics which lead to develop a game

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basics of Graphics and Game	√				
2	Important Aspects of Mathematics and Physics		√			
3	Graphics Programming Fundamentals	√		√		

4	Animation, Transformations and Collision Detection			√		
5	Actual Game Programming				√	
6	Game Engine Introduction					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	1	2	-	2	1	1	-	1	1	2	3
CO2	3	2	-	2	2	-	2	1	1	-	1	1	2	1
CO3	3	1	1	3	2	-	2	1	1	-	2	2	2	1
CO4	3	2	2	2	2	1	2	3	1	2	3	3	3	2
CO5	3	3	3	2	2	1	2	3	1	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS224: Advanced Python Programming
(200 Marks)

Contact Hours: 06

Pre-requisite: Basics of Programming.

Methodology & Pedagogy: Theory sessions are required to address computational power of python through its ability to deploy programs using functional, object oriented and advance python based aspects. Practical sessions demonstrate the implementation of the concepts which are taught during the theory sessions. Case study will help the students to come out with one working module in any of the advanced python programming.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Fundamentals of Python	6	6
2	Iterable Object in Python and Function	7	6
3	Object oriented Programming with Python	5	6
4	GUI Programming using Python	6	6
5	Python libraries and Data collection	6	6
6	Use of Python to develop AI based application	6	6

Total Hours (Theory): 36

Total Hours (Lab) :36

Total Hours:72

Detail Syllabus:

Unit I: Fundamentals of Python

Hours 06

Introduction to Python Program, Features of Python. Environmental setup for Python Programming, Basic program structure to write Python, Data Type, Input and Output Statements, Comment in Python, If Statement with its variation, While and For Loop.

Unit II: Iterable Object in Python and Function

Hours 07

List, Tuple, Set, Dictionary and custom iterable object

What are functions, calling functions, creating functions, passing functions, formal arguments, variable length arguments, default arguments, returning values from the functions, returning multiple values from the functions.

Unit III: Object oriented Programming with Python

Hours 05

Basics of class, object and instance. Class level attribute and instance level attribute. Constructor and other magic methods. Bound and unbound methods. Built in functions for python class and objects

Unit IV: GUI Programming using Python

Hours 06

Introduction, Python's GUI library: Tkinter (installing and getting started), Introduction to GUI programming, top-level window, Tk widgets: Label, Button, scale, scroll bar, text, Checkbutton, Radiobutton. Developing GUI based applications using Tkinter.

Unit V: Python libraries and Data collection

Hours 06

NumPy: Introduction, 2D arrays, Numpy arraya Vs. Python List.

Pandas: Introduction, importing data with pandas, Data frames and Series, pre-processing phase of data analysis, handling missing values in data, formatting data to standardize it and make it consistent, normalizing data

Unit VI: Use of Python for Model Development and Evaluation

Hours 06

ML: Introduction to machine learning- Supervised and Unsupervised Learning, Model development using Linear Regression, Model Visualization, Prediction and Decision Making, Model Evaluation: Over-fitting, Under-fitting and Model Selection

DIP: Introduction to digital image Processing, Basic function to import draw and display images using Python.

Core Books:

1. Wesley J. Chun : Core Python Programming, 2nd edition, Pentice Hall,2006.
2. Megnus Lie Hetland : Beginning Python from novice to professional, 2nd edition, Apress,2009.

Reference Books:

1. Mark Lutz : Programming Python, 4th Edition, O'reilly, 2011 .
2. Dusty Philips: Python 3 Object oriented Programming, PACKT publishing, 2010.
3. Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. <https://developers.google.com/edu/python/> [Google's Python class]
2. <https://www.learnpython.org/> [Learnpython.org]
3. <https://docs.python.org/3/tutorial/> [Python tutorial]
4. <https://www.tutorialspoint.com/python/index.htm> [Tutorials point]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will be able to understand scripting and programming constructs of Python.
CO2 :	Understanding Python's iterable objects and functions.
CO3 :	Clear Understanding of the object-oriented concepts using Python.
CO4 :	Students exposed to GUI programming and Python libraries to handle data.
CO5 :	Students will learn model development using advance concepts of Machine Learning and its applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Fundamentals of Python	√				
2	Iterable Object in Python and Function		√			
3	Object oriented Programming with Python			√		
4	GUI Programming using Python				√	
5	Python libraries and Data collection				√	
6	Use of Python to develop AI based application					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	2	3	2	1	1	1	1	3	2
CO2	2	3	3	3	3	2	3	2	1	1	1	1	3	2
CO3	2	3	3	3	3	2	3	3	2	1	2	1	3	2
CO4	3	3	3	3	3	3	3	3	2	2	3	2	3	2
CO5	3	3	3	3	3	3	3	3	1	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS225: Blockchain Programming
(200 Marks)

Contact Hours: 06

Pre-requisite: None

Methodology & Pedagogy: During the lecture sessions, the students will learn about various sub systems that work in integrated manner to make blockchain work. The teacher will introduce students to concepts of bitcoin and ethereum technologies. The students will also understand about application domains where blockchain can be applied. In the lab sessions, the students will learn to program smart contracts with solidity on ethereum platform.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to blockchain	07	36
2	Functioning of blockchain	07	
3	Bitcoin and related concepts	05	
4	Ethereum and related concepts	09	
5	Introduction to solidity programming	04	
6	Advanced concepts in solidity programming	04	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: Introduction to blockchain

Hours 07

History of blockchain, Centralized, decentralized and distributed systems, Layers of blockchain, Importance of blockchain, Applications of blockchain

Unit II: Functioning of blockchain

Hours 07

Cryptographic concepts- Symmetric key cryptography, Cryptographic hash function, Asymmetric key cryptography, Byzantine general's problem, Merkel tree, Structure of block

Unit III: Bitcoin and related concepts

Hours 05

What is bitcoin? Working with bitcoin, Bitcoin network, Bitcoin mining, Consensus algorithms

Unit IV: Ethereum and related concepts

Hours 09

Design of ethereum, Ethereum blockchain, Smart contracts in ethereum, Ethereum virtual machine, Ethereum eco system

Unit V: Introduction to solidity programming

Hours 04

Ethereum networks, Geth, Solidity compiler, Web3 library, Metamask wallet, emix, Ganache, Structure of ethereum smart contract, datatypes in solidity

Unit VI: Advanced concepts in solidity programming

Hours 04

Global variables and functions, expressions and control structures, writing smart contracts in solidity

Core Books:

9. Bikramaditya Singhal, Gautam Dhameja, Priyansu Sekhar Panda: Beginning Blockchain - A Beginner's Guide to Building Blockchain Solutions: 1st Edition: APress Publication: 2018.
10. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder: Bitcoin and Cryptocurrency Technologies - A Comprehensive Introduction: 1st Edition: Princeton University Press: 2016.
11. Ritesh Modi: Solidity Programming Essentials - A Beginner's Guide to Build Smart Contracts for Ethereum and Blockchain: 1st Edition: Packt Publishing: 2018

Reference Books:

9. Fabian Schar, Aleksander Berentsen: Bitcoin, Blockchain, and Cryptoassets: 1st Edition: MIT Press: 2020
10. Kevin Solorio, Randall Kanna, David H. Hoover: Hands-On Smart Contract Development with Solidity and Ethereum From Fundamentals to Deployment: 1st Edition: O'Reilly Media: 2019

Web References:

7. <https://andersbrownworth.com/cms/460/blockchain/demo> [Blockchain Demo]
8. <https://ethdocs.org/en/latest/introduction/what-is-ethereum.html> [Ethereum Introduction]
9. <https://www.youtube.com/watch?v=M576WGiDBdQ> [Solidity Programming]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	The students will learn about underlying technologies of blockchain and various applications of blockchain technology.
CO2 :	The students will understand working of blockchain in detail.
CO3 :	The students will learn about how the blockchain is used to power bitcoin.
CO4 :	The students will see how ethereum utilizes blockchain.

CO5 :	The students will be able to create and deploy solidity based smart contracts on ethereum platform.
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Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to blockchain	√				
2	Functioning of blockchain		√			
3	Bitcoin and related concepts			√		
4	Ethereum and related concepts				√	
5	Introduction to solidity programming					√
6	Advanced concepts in solidity programming					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	1	3	1	2	1	2	-	-	1	1	-
CO2	3	-	2	2	2	1	1	2	1	-	-	1	2	-
CO3	2	2	3	3	3	2	2	1	2	-	-	1	2	-
CO4	3	-	3	3	3	2	3	3	1	-	1	2	3	-
CO5	3	-	3	3	3	2	3	3	2	-	2	3	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS226: HTTP Web Service for Enterprise Application

(200 Marks)

Contact Hours: 06

Pre-requisite: Work experience with C#, ASP.NET MVC, MSSQL, HTML, CSS, JavaScript and have some understanding of JQuery.

Methodology & Pedagogy: During theory lectures, illustrations emphasizing the need for advance features of WEB API and ASP.NET will be covered. During Practical sessions, students will require to develop Web API using concepts of .NET framework and other pre-requisite technologies, discussed during class.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	ASP.NET Web API - Overview	4	36
2	MVC and Web API Controller with Entity Framework	8	
3	Web API Routing	4	
4	Web API (Action Formatter and Filters)	4	
5	Backend Validation with Exception Handling	8	
6	AJAX and JQuery	8	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: ASP.NET Web API - Overview

Hours 04

Introduction to RESTful WEB API, Characteristics of ASP.NET WEB API, versions of Web API, Difference among web service, Window Communication Foundation and WEB API, Test Web API Fiddler and Postman

Unit II: MVC and Web API Controller with Entity Framework

Hours 08

Use of Web API Controller in Controller class of Web API, Functionality of Web API Controller, Difference between Web API controller and MVC controller, Action Method Naming Conventions, Action

Result, MVC with WEB API, CRUD operation with Entity Framework in Web API.

Unit III: Web API Routing

Hours 04

Routing in Web API, Routing: Convention-based Routing and Attribute based Routing, Routing and Action Execution.

Unit IV: Web API (Action Formatter and Filters)

Hours 04

Data Formatter, Media Type Formatter, Web API Filter, Exception Filters: HttpResponseException, Exception Filters, Registering Exception Filters, HttpError

Unit V: Backend Validation with Exception Handling

Hours 08

Model Validation with annotation, Custom Exception, recognizing need of custom Exceptions, Backend validation using Custom Exception for robustness and Data Integrity.

Unit VI: AJAX and JQuery

Hours 08

Consume RESTful Web API through GET, POST, PUT and DELETE. Understanding JQuery and JavaScript. JQuery usage and AJAX Request configuration and parameters understanding.

Core Books:

12. Mithun Pattankar, Malendra Hurbuns, Mastering ASP.NET Web API: Build powerful HTTP services and make the most of the ASP.NET Core Web API platform 1st Edition. – 2017.

Reference Books:

11. Jamie Kurtz, Brian Wortman, ASP.NET Web API 2: Building a REST Service from Start to Finish. - 2014

Web References:

1. https://www.tutorialspoint.com/asp.net_mvc/asp.net_mvc_web_api.htm

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be familiar with RESTful API and ASP .NET Web API.
CO2 :	Be able to implement ASP .NET Web API Controller and configure its Routing and implementing Entity Framework.
CO3 :	Be able to Configure actions in API Controllers, Filters and Formatters.
CO4 :	Be able provide Data Integrity with validations with Exceptions.
CO5 :	Be able to call API through testing applications and AJAX calls from 3 rd Party Applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	ASP.NET Web API - Overview	√				
2	MVC and Web API Controller with Entity Framework		√			
3	Web API – Routing		√			
4	Web API (Action Formatter and Filters)			√		
5	Backend Validation with Exception Handling				√	
6	AJAX and JQuery					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	3	3	1	3	-	-	2	2	2	2	2	2	2
CO2	2	1	2	2	3	2	2	3	-	3	2	2	3	3
CO3	2	2	2	2	3	2	2	2	-	1	2	2	3	2
CO4	3	3	3	3	2	3	2	2	-	2	3	2	3	2
CO5	2	3	3	1	3	2	2	3	2	2	3	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS227: Web Development using Open Source
(200 Marks)

Contact Hours: 06

Pre-requisite: Basic understanding of HTML and MySQL

Methodology & Pedagogy: During theory and practical sessions, students able to install & configure PHP and prerequisite software(s). Also, student will be emphasized to develop dynamic web applications.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Open Source Software and PHP	07	36
2	Basics of PHP	06	
3	Array and OOP in PHP	05	
4	Form Handling using PHP	05	
5	PHP Utilities	06	
6	Relational Databases using PHP	07	

Total Hours (Theory):36

Total Hours (Lab) :36

Total Hours:72

Detail Syllabus:

Unit I : Introduction to Open Source Software and PHP

Hours 07

Overview of Open Source Software, Widely used Open Source Products, Development Philosophy, Open source vs. Closed Source, Open Source Technology Importance, Installation & Configuration of PHP, Introduction to PHP, Working of HTML with PHP

Unit II : Basics of PHP

Hours 06

PHP language Basics: Lexical Structure, Data types, Variables, Expressions and Operators, Control and Looping statements. Functions: Function Definition, Function Parameters, Returning Values. Strings: Usages and String Functions

Unit III : Array and OOP in PHP**Hours 05**

Arrays: Types of Arrays and its Usages, Array functions. Objects: Declaring Class, Properties, Methods, Exception Handling, Examples.

Unit IV : Form Handling using PHP**Hours 05**

Capturing data with PHP Using HTML Form Elements, Send Form data using GET Method & POST Method, Receive Form data using \$_GET, \$_POST & \$_REQUEST variables, Super Global Variables.

Unit V : PHP Utilities**Hours 06**

File Uploading: Upload Single and Multiple file using PHP script, Understanding HTTP requests, Exploring and modifying HTTP responses, getting information from web server, Sending mails

Unit VI : Relational Databases using PHP**Hours 07**

Relational Databases and SQL, Using PHP to access Databases, PHP Data Objects (PDO), CRUD operations, Handling Errors, State Management Techniques: Concept of Session, starting session, modifying session variables, Un registering and deleting session variable, Concept of Cookies

Core Books:

13. Vikram Vaswani, PHP: A Beginner's Guide: Indian Edition, First Edition, McGraw Hill, 2009
14. Matt Doyle , Beginning PHP 5.3 , Wrox , 2010

Reference Books:

12. Timothy Boronczyk , Elizabeth Naramore, Jason Gerner, Yann Le Scouarnec, Jeremy Stolz, Michael K. Glass): Beginning PHP6, Apache, MySQL Web Development : Wrox , 2009.
13. Lynn Beighley and Michael Morrison, Head First PHP & MySQL, First Edition, O'Reilly Publication, 2009

Web References:

10. <https://github.com/PHPMailer/PHPMailer> [PHPMailer Code]
11. <https://www.php.net/> [Official website of PHP]
12. <https://www.geeksforgeeks.org/php-tutorials/> [Lecture notes of PHP]
13. <https://www.w3schools.com/php/default.asp> [Lecture notes of PHP]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Gain understanding of Open Source software and PHP.
CO2 :	To utilize knowledge and skills for basics of PHP and functions of PHP
CO3 :	Acquire the knowledge of array and OOP in PHP.
CO4 :	Learn form handling and utilities in PHP.
CO5 :	Be able to develop dynamic web based application using PHP and MySQL with state management techniques.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Open Source Software and PHP	√				
2	Basics of PHP		√			
3	Array and OOP in PHP			√		
4	Form Handling using PHP				√	
5	PHP Utilities				√	
6	Relational Databases using PHP					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	3	-	1	-	1	2	-	-	2	-	3
CO2	3	3	3	3	-	1	-	1	1	-	-	3	-	3
CO3	3	3	3	3	-	1	-	1	1	-	-	3	-	3
CO4	3	3	3	3	-	1	-	1	1	-	-	3	-	3
CO5	3	3	3	3	2	1	1	1	2	1	1	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS228: Advanced Mobile Application Development
(200 Marks)

Contact Hours: 06

Pre-requisite: Object Oriented Programming Concepts

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for basic features of Mobile Computing and Cross Platform- the Mobile Application Development platform will be given. During Practical sessions, students will be required to develop Mobile Application using Dart language in Flutter. Student shall also develop applications with elegant user interface that deal with data storage using Firebase and state management

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Cross-platform advance with Flutter & Dart	05	36
2	UI Designing with Flutter	07	
3	State management with App creation	05	
4	Architecting Flutter applications and Its packages	08	
5	Introduction to Backend to Flutter-I	05	
6	Introduction to Backend to Flutter-II	06	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: Introduction to Cross-platform advance with Flutter & Dart

Hours 05

Set up a new Flutter project using Android Studio. Widget tree, Interface design: pre-made Flutter Widgets. Image and Text Widgets. App Icons for iOS and Android. Add and load image assets to Flutter projects. Run Flutter apps on iOS Simulator, Android Emulator and physical devices.

Unit II: UI Designing with Flutter

Hours 07

Hot Reload and Hot Restart, Use of Pubspec.yaml file, custom assets and fonts. An introduction to the Widget build(), layout widgets: Columns, Rows, Containers and Cards. Material icons, Icons class. Theme widgets. Refactoring widgets. Dart annotations and modifiers. Immutability of Stateless and Stateful Widgets. Update screen with the build(). Custom Flutter Widgets. Difference between final and const in Dart. Maps, enums and the ternary operator. Functions and arguments in Dart. Multi-screen Flutter apps, routes and the Navigator widget. Flutter favours composition vs. inheritance (customizing widgets).

Unit III: State management with App creation

Hours 05

About Stateful and Stateless Widgets, callbacks. Declarative style of UI programming, Flutter widgets react to state changes. Import Dart libraries. Variables, data types and functions in Dart. Build flexible layouts. Understand the relationship between setState(), State objects and Stateful Widgets.

Unit IV: Architecting Flutter applications and Its packages

Hours 08

Dart package manager, Flutter compatible packages. The structure of the pubspec.yaml file. Incorporate the audio players package to play sound. Functions in Dart and arrow syntax. Refactor widgets, Flutter's philosophy of UI as code. The lists and conditionals in Dart. Classes and objects. Understand Object Oriented Dart. Dart Constructors for Flutter widgets. Design patterns to structure Flutter apps. Structuring and organizing Flutter apps.

Unit V: Introduction to Backend to Flutter-I

Hours 05

Asynchronous programming in Dart and use of async/await. Stateful Widget lifecycle methods, Handling exceptions in dart. Null aware operators. Location data from both iOS and Android. Http package and live data from open APIs. Parse JSON data using the dart:convert library.

Unit VI: Firebase and State Management with Flutter-II

Hours 06

State objects via the Stateful Widget. Use the TextField Widget to take user input. Pass data backwards using the Navigator widget. Hero animations in Flutter apps. Animation controller, custom animations. The use of Dart mixins. Firebase Cloud Firestore into your Flutter apps. Authentication with Firebase Auth package.

Core Books:

1. Marco L. Napoli: Beginning Flutter: A Hands On Guide to App Development: Wrox publication: 2019.

Reference Books:

1. Eric Windmill: Flutter in Action: Edition: Manning Publication: January 2020.
2. Alessandro Biessek: Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter and Dart 2: Packt publication: September 2019

Web References:

1. <https://docs.flutter.dev/reference/tutorials>
2. <https://www.tutorialspoint.com/flutter/index.htm>
3. <https://www.javatpoint.com/flutter>
4. <https://fluttertutorial.in/>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 : able to clear all object oriented programming and cross platform concepts

CO2 :	able to learn Flutter and Dart step by step
CO3 :	able to learn the reduce the code through native app performance, animated UI with material design and least testing
CO4 :	able to use Firebase to authenticate the users and use the remote database
CO5 :	able to build engaging native mobile apps for both Android and iOS

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to cross-platform development with Flutter and Dart	√				
2	Creating UI with Flutter	√	√			
3	Building Apps with state			√		
4	Leveraging Flutter packages and structuring Flutter applications			√		
5	Incorporating backend data with Flutter application					√
6	Integrating Flutter Application with Firebase and State Management				√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	3	2	1	3	-	1	-	1	-	-	1
CO2	3	1	3	2	3	1	2	1	1	-	1	1	1	2
CO3	3	-	3	1	3	1	1	2	1	-	1	1	1	2
CO4	3	3	3	3	3	1	1	3	1	-	2	2	3	1
CO5	3	3	3	1	3	1	1	3	1	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS229: Modern Software Engineering
(100 Marks)

Contact Hours: 04 hours

Pre-requisite: Analysis and design of Information Systems, Fundamental concepts of Object Oriented Programming.

Methodology & Pedagogy: During theory sessions, the focus will be given on different concepts of software engineering and related aspects such as Project management and Agile software development. The emphasis will be given on the basics of software engineering concepts and UML. Students will get exposure to various concepts of DevOps, as well as Process Improvement and Reengineering.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Software Engineering	7
2	Requirement analysis and project management	8
3	Understanding UML	8
4	Agile Development: XP and SCRUM	9
5	DevOps	8
6	Software maintenance and risk management	8

Total Hours (Theory): 48

Total Hours: 48

Detail Syllabus:

Unit I: Introduction to Software Engineering

Hours 07

The Role of Software in today's era, Software Engineering: A Layered Technology/software process layers, Software Process Models, The Linear Sequential Model, The Prototyping Model, Evolutionary Process Models, Introduction to software process: Generic process framework activities, Umbrella activities.

Unit II Requirement analysis and project management

Hours 08

Understanding the Requirement, Requirement Modeling, Requirement Specification (SRS), Requirement Analysis and Requirement Elicitation, Requirement Engineering.

Introduction to the basic concepts of Project Scheduling, Project scheduling principles, relationships and effort distribution, Process Definition and Tailoring, Process Database and Process Capability Baseline, Effort Estimation and Scheduling, Project Management Plan, Configuration Management.

Unit III: Understanding UML

Hours 08

Introduction to UML, Structure modelling: Class Diagram and Behavioral Modelling. Under Behavioral Modelling: Use case Diagram, Interaction Diagram, Activity Diagram.

Unit IV: Agile Development

Hours 09

Introduction to Agile Software Development, Characteristics of Agile Process, Agile methods, Principles of Agile methods, Problems with Agile methods, Extreme Programming, The Four Core Values of XP.

Unit V: Overview of DevOps

Hours 08

Overview, Problem Case Definition, Benefits of Fixing Application Development Challenges, DevOps Adoption Approach through Assessment, Solution Dimensions, What is DevOps?, DevOps Importance and Benefits, DevOps Principles and Practices, 7 C's of DevOps Lifecycle for Business Agility, How to Choose Right DevOps Tools, Challenges with DevOps Implementation, Must Do Things for DevOps.

Unit VI: Software maintenance and risk management

Hours 08

Software risks, risk identification, risk projection, risk mitigation, monitoring and management. Introduction to software maintenance, types of maintenance.

Concept of software reengineering, Business Process Reengineering(BPR), BPR life cycle, reverse engineering, Applications of Software Engineering.

Core Books:

1. Roger S.Pressman, Software Engineering- A practitioner's Approach, McGraw-Hill International Editions
2. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User
3. Rajib Mall, Fundamentals of software Engineering, Prentice Hall of India.

Reference Books:

1. Ian Sommerville, Software engineering, Pearson education Asia
2. Pankaj Jalote, Software Engineering – A Precise Approach Wiley
3. Merlin Dorfman (Editor), Richard H. Thayer (Editor) ,Software Engineering
4. Robert C. "Uncle Bob" Martin , Clean Architecture: A Craftsman's Guide to Software Structure and Design
5. Deepak Gaikwad, Viral Thakkar, DevOps Tools from Practitioner's ViewPoint, Wiley

Web References:

1. https://en.wikibooks.org/wiki/Introduction_to_Software_Engineering/Process/Methodology [Introduction to Software Engineering/Process/Methodology]
2. <https://www.uml-diagrams.org/> [UMLUnits]
3. <https://nptel.ac.in/courses/106/101/106101061/> [Agile Software Development and Extreme Programming and Overview of DevOps]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understanding Software Engineering Process Models
CO2 :	Able to analyze requirements and manage project
CO3 :	Get familiar with the concepts of Unified Modeling Language
CO4 :	Understanding of the Agile software development concepts
CO5 :	Able to apply DevOps concepts and understand risk management

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Software Engineering	√				
2	Requirement analysis and project management		√			
3	Understanding UML			√		
4	Agile Development				√	
5	DevOps					√
6	Software maintenance and risk management					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	3	3	2	2	2	1	2	3	3
CO2	3	2	2	2	2	3	3	3	3	2	3	2	3	3
CO3	2	2	3	2	2	2	2	2	2	1	1	2	3	2
CO4	3	3	3	2	2	2	2	1	1	1	3	2	3	2
CO5	2	2	3	2	1	3	2	1	1	1	2	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS106.02 C: ACADEMIC WRITING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

- An Intermediate Guide to Writing in English for University Study
<https://www.futurelearn.com/courses/english-for-study-intermediate/4/todo/62943>

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Academic Writing and Research Process	05
2.	Anatomy of Academic Writing	05
3.	Key Academic Skills	05
4.	Accuracy in Academic Writing	05
5.	Using and Citing Sources of Ideas	05
6.	Contemporary Practices in Academic Writing	05

Total hours (Practical): 30

Total hours (Lab): --

Total hours :30

Detailed Syllabus:

1.	Academic Writing and Research Process	5 Hours
	Introduction to Academic Writing, Academic Writing as a Part of Research, Types of Academic Writing, Features of Academic Writing, Importance of Good Academic Writing in various Academic Works	
2.	Anatomy of Academic Writing	5 Hours
	Academic Vocabulary, Simple and Complex Sentences, Organizing Paragraphs, The Writing Process, Adopting Academic Writing Style	

3.	Key Academic Skills	5 Hours
	Note – taking, Note – making, Paraphrasing, Summarizing	
4.	Accuracy in Academic Writing	5 Hours
	Lexical Range, Academic Language and Structures, Elements of Writing, Proof Reading, Editing, and Rewriting	
5.	Using and Citing Sources of Ideas	5 Hours
	Academic Texts and their Types, Intellectual Honesty in Academic Writing, Avoiding Plagiarism – Idea Theft, Degrees of Plagiarism, Types of Borrowing, Anatomy of Citations, Common Citation Styles	
6.	Contemporary Practices in Academic Writing	5 Hours
	Analytical Essays, Graph / Table / Process Interpretation and Description, Writing Reports and Abstract, Writing Research / Concept Papers	

Course Outcome (COs):

At the end of the course, the students

CO1	Will have sound understanding of the concept and applications of academic writing
CO2	Will have acquired enough knowledge of academic writing style, strategy and approach
CO3	Will be able to demonstrate error free and effective academic writing
CO4	Will be able to demonstrate ability to work on project/report/paper writing
CO5	Will have the sound understanding of the Research and Research Methodology
CO6	Will be effectively communicating in diverse academic and professional settings.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	3	-	3	-	-	-	-	2
CO6	-	-	-	-	-	-	2	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

Text book:

1. Academic Writing for International Students, Routledge
2. Academic Writing: A Guide for Management Students and Researchers. Monipally, M.M. &Pawar, B.S. Sage. 2010. New Delhi
3. Effective Academic Writing Level - 1,2,3,4 (Second Edition) By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; Publisher: Oxford

Reference book:

1. Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
2. Development Communication In Practice by Vilanilam V J, Sage
3. Intercultural Communication by Mingsheng Li, Patel Fay, Sage

Web material:

www.owl.perdue.edu